

Cultural Differences in Indirect Speech Act Use and Politeness in Human–Robot Interaction

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Abstract—How do native English speakers and native Korean speakers politely make a request to a robot? Previous human–robot interaction studies on English have demonstrated that humans use indirect speech acts (ISAs) frequently to robots to make their requests polite. However, it is unknown whether humans considerably used ISAs to robots in other languages. In addition to ISAs, Korean has other politeness expressions called honorifics, which indicate different politeness from that of ISAs. This study aimed to investigate the cultural differences in humans’ politeness expressions and politeness when they make requests to robots and to re-examine the effect of conventionality of context on the use of politeness expressions. We conducted a replication experiment of Williams et al. (2018) on native Korean speakers and analyzed their use of ISAs and honorifics. Our results showed that ISAs are rarely used in task-based human–robot interaction in Korean. Instead, honorifics are more frequently used than ISAs and are more common in conventionalized contexts than in unconventionalized contexts. These results suggest that the difference in politeness expressions and politeness between English and Korean exist in both human–robot interaction and human–human interaction. Furthermore, the conventionality of context has a strong constraint on making humans follow social norms in both languages.

Keywords—Cultural differences, Human–robot interaction, Politeness, Indirect speech acts, Honorifics

I. INTRODUCTION

In human–human conversation, how to say is as important as what to say. For example, in a normal context in English, the imperative “Close the door” can be considered inappropriate, whereas the interrogative “Would you mind closing the door?” can be considered polite. The latter is called an indirect speech act (ISA), an utterance that indirectly performs a speech act by performing another speech act [1]. An ISA is a representative politeness expression for acquiring politeness in human–human interaction (HHI). According to politeness theory [2], a speaker use ISAs to save a public self-image, or a face, of listener.

Recent studies have found that humans frequently use ISAs even to robots in a simple task-based human–robot interaction (HRI) [3], [4]. These studies claimed that humans’ ISA use in HRI is evidence that humans apply the social norm of politeness to HRI, and that understanding the underlying intentions in ISAs is important in natural language understanding (NLU).

However, linguistic studies have argued that there are various politeness expressions other than ISAs, and that different politeness expressions can realize different types of politeness according to the language [5], [6], [7]. For instance, in Korean, besides politeness for saving a face, which is expressed by assertion-softening/reinforcing measures, such as ISAs, there is politeness for indexing social relationships, which is encoded by grammatical and lexical forms, called honorifics. Korean speech level (or speech style) is a grammatical system of Korean honorifics, comprising six morphologically differentiated levels, often re-grouped into two levels: honorific speech level (“contaysmal” in Korean^a) and non-honorific speech level (“panmal” in Korean) [8], [9], [10]. In Korean, the politeness for social indexing is emphasized more than that for face-saving [7], [11]. Therefore, selecting speech level is a necessary condition for conversation, while choosing either ISAs or direct speech acts (DSAs) is a sufficient condition for conversation. If such cultural differences in politeness in HHI affect HRI, differences in the use of politeness expressions will also appear in HRI.

HRI studies examining the effects of a robot’s ISAs have considered cultural differences [12], [13], but those examining the use of human’s ISAs have been mainly conducted with English users [3], [4]. Given the different politeness expressions and politeness depending on language, examining the differences in the use of politeness expressions in HRI among different languages can reveal more detailed social norms in HRI. In addition, without observing the linguistic behaviors of various cultures in HRI, it would be difficult to implement a dialogue system that can adapt appropriately to different cultures. Thus, in this study, we examined how Koreans use politeness expressions, such as ISAs and speech levels (honorifics), in HRI and compared it with the ISA use of English users in HRI by replicating a previous study [4].

^a All Korean expressions and examples in this paper follow Yale Romanization.

II. RELATED WORK

A. Cultural Differences in ISA Use in HRI

Previous HRI studies have investigated robots' ISAs, which affect humans' robot perception, or humans' ISAs, which need to be processed in NLU. Based on ISA use in HHI, both types of HRI studies have mainly investigated requests that tend to be formalized as ISAs in HHI or considered politeness related to ISA as a social norm that can be realized in HRI. However, despite the inherent cultural differences in ISA use in HHI, few studies have focused on ISA use in HRI in terms of cultural differences.

Previous studies on robots' ISAs have demonstrated that they have positive effects on human perception of robots. Saunderson and Nejat [12] suggested that a robot's indirect requests are perceived to be more trustworthy and encourage greater willingness to help the robot than the robot's direct requests. However, Salem et al. [13] demonstrated that differences in the use of politeness strategies between Arabic and English may influence the robot perceptions of speakers of each language. When the robot's speech was changed in such a way using a positive politeness strategy (highlighting common ground (e.g., "let's close the window, please")) instead of using bald on record strategy (expressing directly: "close the window"), the closeness of the robot to female Arabian participants was decreased as opposed to the female English participants.

Previous studies on humans' ISAs, which are topics of our interest, have examined frequency of humans' ISAs and showed that humans use numerous ISAs when making requests to robots in simple tasks. Briggs et al. [3] demonstrated that approximately 28% of all task-related utterances were produced as ISAs when humans requested robots to destroy a tower. Williams et al. [4] also observed that the median rates of ISA use were 16% for the same tower-destroying task of Briggs et al. [3] and 75% for the food-ordering task, where humans gave orders to robots just as in a restaurant. This suggests that conventionality of context affects humans' ISA use in HRI. Both studies claimed that utterance errors caused by an inability to understand ISAs can act as severe obstacles in NLU, as ISA use rates were higher than the word error rates on the Switchboard Corpus [14]. Furthermore, both studies have suggested that humans apply social norms, such as politeness, to robots. However, these observations were limited to one language (i.e., English). To the best of our knowledge, there are no cross-cultural studies or studies on other languages about humans' ISAs that have focused on cultural and linguistic differences, unlike the studies on robots' ISAs.

In a previous study conducted in Korean with a smart speaker, Kim [15] observed that the rate of ISA use was as low as 4.2% when requesting information. This result suggests that ISA use in HRI and the importance of a robot's ability to understand ISAs differ across languages. However, it is difficult to directly compare the results of Briggs et al. [3] and Williams et al. [4] with those of Kim [15], as these studies used different agents (a robot vs. a smart speaker) and different

requesting contexts (requesting action vs. requesting information). Thus, here, we replicated the experiment conducted by Williams et al. [4] to compare the differences in ISA use in HRI between English and Korean.

B. Politeness Strategies Across Languages

Many linguistic studies have mentioned that the use of linguistic indirectness to achieve politeness is universal [2]. However, indirectness can be considered a relatively less important politeness device, depending on the language. For example, in Russian, an imperative form is common and appropriate in most request contexts, unlike English [16]. House and Kasper [17] found that German speakers use direct forms more frequently than English speakers.

The relatively low importance of acquiring politeness in some languages can be attributed to the existence of politeness expressions other than ISAs. Politeness expressions can be divided into several types according to the different aspects of a language. For example, an ISA is a pragmatic expression, an address term (e.g., "Sir" and "Ma'am") is a lexical expression, and Korean speech level using different sentence-final endings (e.g., "supnita" in "komawe-supnita" ("thanked (you)")) is a grammatical expression. Also, these different types of politeness expressions may indicate different politeness. For example, there are volitional politeness and discernment politeness [6]. Volitional politeness is used individually to minimize face-threatening behavior and is influenced by interactive speech act situations. Conversely, discernment politeness is used obligatorily to index interpersonal relationships and is determined by the cultural norms of a society. While volitional politeness is often realized using a pragmatic expression, discernment politeness is often realized using lexical and grammatical expressions. For example, Byon [7] demonstrated that different speech levels were selected according to social relationships rather than adjusting the degree of indirectness. In addition, Korean speakers more frequently used DSAs than ISAs across all situations. This suggests that discernment politeness plays a more important role than volitional politeness in the Korean linguistic politeness system.

We assume that these cross-linguistic differences in politeness expressions and politeness manifested in HHI will also be reflected in HRI. Thus, while comparing the use of ISA between English and Korean, we examined the use of speech level (honorific), which is an important politeness expression in Korean, to determine how the politeness norm is realized in HRI differently depending on the language. Furthermore, we re-examined the effect of conventionality of context on the use of politeness expressions in HRI, which was manifested in Williams et al. [4].

III. HYPOTHESES

Based on related works on cultural differences in ISA use in HRI and politeness strategies across different languages, we constructed five hypotheses for our experiments.

H1. Compared to Williams et al. [4], who used English, the overall frequency of ISA use will be lower in Korean.

H2. A low frequency of ISA use will occur regardless of the robot's ability to understand ISAs.

H3. A low frequency of ISA use will occur in both conventional and unconventional task contexts.

H4. A robot's ability to understand ISAs will not affect a human's perception of the robot. As the frequency of ISA use will be low regardless of the robot's ability to understand ISAs, the robot will fulfill almost all human requests irrespective of its ability to understand ISAs.

H5. Different frequencies of honorific use will occur between conventional and unconventional task contexts. In the conventional context, the frequency of honorific use will be higher.

IV. METHODS

We conducted a Wizard-of-Oz (WOZ) between-subject experiment replicating Williams et al.'s [4] experiment.

A. Participants

A total of 48 native Korean speakers (24 men, 24 women; $M_{age} = 26.15$, $SD_{age} = 3.57$) living in Korea participated in the experiment. The participants were recruited through emails and online advertisements at the Korea Institute of Science and Technology (KIST IRB No. 2020-027). They were undergraduate or graduate students but did not major in HRI or human-computer interaction (HCI). Following Williams et al. [4], we asked them for information regarding their majors and robot experiences. The comparisons between our study's participants and Williams et al.'s [4] participants are summarized in Table I. Each participant was paid approximately \$9 for the experiment.

TABLE I. COMPARISON OF PARTICIPANT INFORMATION

	Williams et al. [4]	Our study
Language	English	Korean
Age	19–69 (34.1 ± 16.2)	21–38 (26.15 ± 3.57)
Gender	21 men, 28 women	24 men, 24 women
Major	Mathematics, Computer science, or Engineering: 6 Natural science or Medicine: 15 Social science: 10 Arts or Humanities: 7 Other field: 5 No major: 0	Computer science or Engineering: 29 Natural science: 12 Integrated science: 7
Robot experience	Having seen a robot in a movie: 92% (45/49) Having seen a robot in real life: 45% (22/49) Having interacted with a robot before: 22% (11/49)	Having seen a robot in a movie: 96% (46/48) Having seen a robot in real life: 73% (35/48) Having interacted with a robot before: 50% (24/48)

B. Experimental Conditions

Following Williams et al.'s [4] experimental conditions, the participants were evenly distributed into four experimental conditions according to the task context (context condition) and the robot's ability to understand ISAs (dialogue condition) (Fig. 2). The context condition comprised a restaurant (REST) scenario and a demolition (DEMO) scenario. In the REST scenario, the participants requested a robot to serve food; in the DEMO scenario, they requested a robot to topple a tower. Williams et al. [4] suggested that the REST scenario is a conventionalized context in which participants must have experienced in giving orders to human waiters, while the DEMO scenario is an unconventionalized context to which participants would be unaccustomed. In contrast, the dialogue condition comprised an understanding (UND) dialogue and a misunderstanding (MIS) dialogue. In the UND dialogue, if the utterances of the participants were task-related, the robot answered "Yes, all right." ("Ney, alkeyssupnita." in Korean) and executed the participants' requests, regardless of whether they were ISAs or not. However, in the MIS dialogue, when the requests were ISAs, the robot answered literally (e.g., for the request "Can you give me some water?," it answered "Yes, I can do it. Please tell me your order." ("Ney, hal swu issupnita. Cwumwunhay cwuseyyo." in Korean) and made no movement, but when the requests were not ISAs, the robot answered, "Yes, all right." and fulfilled the requests in the same way as in the case of the UND dialogue. In addition, for the MIS dialogue, the participants could make a request again after a failure caused by ISA. The robot did not move until the participants finally produced non-ISAs. Under both conditions, all of the robot's responses were controlled as honorifics.

C. Experimental Setup

We used Turtlebot with Raspberry Pi and a small Bluetooth speaker (Yoitch YS-BS400) instead of iRobot Create used by Williams et al. [4] (Fig. 1). The webcam was installed with a tripod in a corner of the experimental room to observe the robot's movement. The robot's movement was controlled by a Bluetooth keyboard (Logitech K345), and its speech was played by selecting a prerecorded audio file (ko-KR-Wavenet-D voice from Google TTS) on a notebook (ThinkPad X1 Extreme), which was connected to the Bluetooth speaker. For the REST scenario, we used three laminated paper cards with a food picture (water, cake, and noodle) and plastic cups of three different colors (pink, yellow, and sky blue) to make the towers for the DEMO scenario. In the REST scenario, the room was empty; in the DEMO scenario, the room had three colored towers of plastic cups placed in front of the participants.

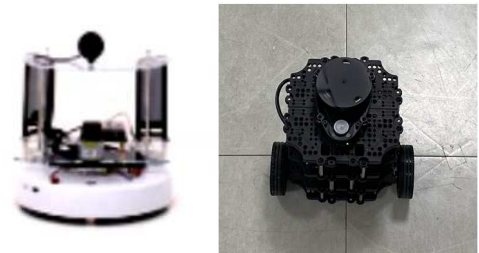


Fig. 1. iRobot Create in Williams et al. (2018) (left) and Turtlebot in our experiment (right)

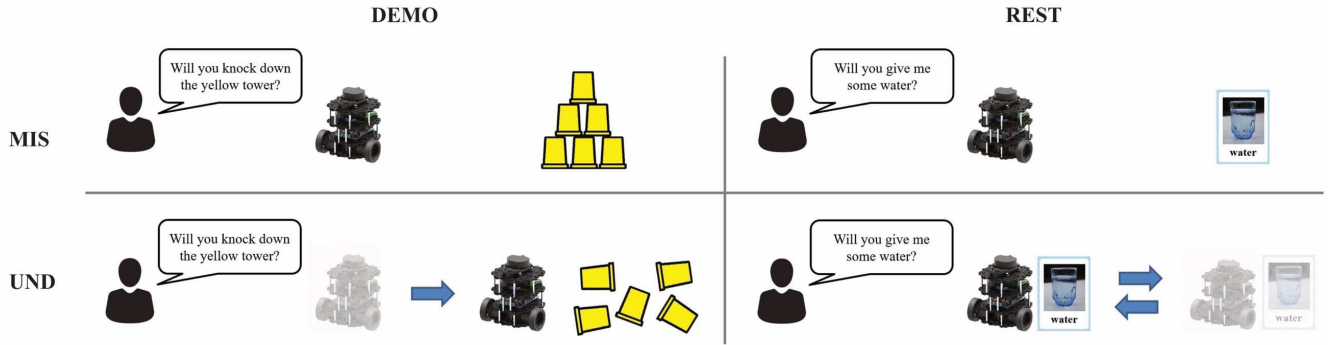


Fig. 2. Experimental conditions.

D. Procedures

Upon arrival in the laboratory, the participants read and signed a written informed consent form. For all experimental conditions, the experimenter gave the instruction to the participants that the experiment was designed to develop the natural language ability of the robot. After the instruction, the participants were provided with a list of three foods or three towers depending on the context condition. Referring to the list, the participants made three requests one-by-one in no specific order to the robot, whose job was to serve food in the REST scenario or topple towers in the DEMO scenario. At first, the robot showed up from an opaque partition behind the participants and introduced itself in front of them, saying, “Hello, my name is Waiterbot/Towerbot. What can I do for you?” After the participants made a request (e.g., “bring water” (REST) or “knock down the yellow tower” (DEMO)), the robot fulfilled the request (e.g., brought the card with the water picture (REST) or rushed to the yellow tower and knocked it down (DEMO)), and the same interaction continued for two more times. During the experiment, all verbal interactions were recorded using Tascam DR-22WL. After the interaction, the participants answered a post-questionnaire about their perceptions of the robot through the Google survey forms displayed on the tablet computer (Apple iPad 6).

E. Measures

This study used behavioral measures for assessing **H1–H3** and **H5** and subjective measures for assessing **H4**. For the behavioral measures, all participant requests made in the experiment were orthographically transcribed in Korean by listening to the recorded files repeatedly. We also transcribed the intonation of the final syllable of sentence-final endings with symbols. The falling tone was marked with a period (“.”), while the rising tone was marked with a question mark (“?”). Then, these transcriptions were annotated according to data analysis criteria for ISAs and honorifics. The subjective measures were the post-questionnaire of 10 questions from Williams et al. [4], translated into Korean with a 10-point Likert scale, which assessed the participants’ perceptions of the robot.

F. Data Analysis

For behavioral measures, we employed the following ISA and honorific judgment criteria presented in Korean linguistic studies. We analyzed task-relevant utterances for the requests made by the participants.

1) ISA Judgement

We made judgments about ISAs by utilizing sentence-final elements. In Korean, sentence-final elements generally determine a sentence type that is conventionally related to a certain speech act (Table II) [18]. There are four sentence types in Korean [8], and their determination usually depends on the morphological forms of sentence-final endings, intonations of their final syllable, and subjects of sentences [19]. For example, if the sentence has the ending “-E” with a rising intonation, its sentence type is an interrogative and its conventionally paired speech act is a question. However, in a context where the speaker asks the listener to do something, the actual speech act of this sentence can be a request. This is similar to “Can you give me water?” in English, which literally asks the listener’s ability to give water or can be a request for water depending on the context. As in the given situation, the participants requested the robot to destroy a tower or bring food, all sentence types that were not imperative were judged as ISAs. Furthermore, if the utterances were noun phrases, we judged them as DSAs as they directly request by indicating objects.

TABLE II. CLASSIFICATION OF SENTENCE TYPE IN KOREAN

Sentence type (Speech acts)	Sentence-final ending ^a	Intonation	Subject
Declarative (Statement)	-(su)pnita , -(u)lkeyyo , -(n)ta , -ney , -(u)lkey	-	-
	-Eyo ^b , -ciyo/-cyo , -(u)llaeyo , -(s)o , -(u)o , -E , -ci , -(u)llae	Falling	Speaker or third party
Interrogative (Question)	-(su)pnikka , -na/-nunka , ni/-nu)nya	-	-
	-Eyo , -ciyo/-cyo , -(u)llaeyo , -(s)o , -(u)o , -E , -ci , -(u)llae	Rising	-
Imperative (Request)	-sipsio , -key , -kela/-Ela	-	-
	-Eyo , -ciyo/-cyo , -(s)o , -(u)o , -E , -ci	Falling	Listener
Propositive (Proposal)	-(u)sipsita , -(u)psita , -sey , -ca	-	-
	-Eyo , -ciyo/-cyo , -E , -ci	Falling	Speaker and listener

^a. Only representative sentence-final endings are shown. Endings found in our data are shown in bold.

^b. -E indicates the alternation between -a and -e [8].

TABLE III. HONORIFIC AND NON-HONORIFIC CLASSIFICATION

Honorific	Speech level	Sentence-final ending ^a
Honorific	Deferential	-(su)pnita , -(su)pnikka, -sipsio, -(u)sipsita
	Blunt	-(s)o, -(u)o, -(u)psita
	Polite	-Eyo^b , -ciyo/-cyo, -(u)llaeyo , -(u)lkeyyo
Non-honorific	Familiar	-ney, -na/-nunka, -key, -sey
	Plain	-(n)ta, -ni/-(nu)nya, -kela/-Ela, -ca
	Intimate	-E , -ci, -(u)llae , -(u)lkey

^a. Only representative sentence-final endings are shown. Endings found in our data are shown in bold.

^b. -E indicates the alternation between -a and -e [8].

2) Honorific Judgement

We made judgments about honorifics by using the sentence-final endings related to the speech levels. Table III shows the honorific/non-honorific classification of speech levels, six speech levels, and the sentence-final endings for each speech level broadly accepted in Korean linguistic studies [8], [20], [21]. For example, utterances ending with “-(su)pnita” or “-Eyo” were judged as honorific, while those ending with “-(n)ta” or “-E” were judged as non-honorific. Noun phrases without sentence final endings were also judged as non-honorific.

V. RESULTS

We used the Shapiro–Wilk test to test the normality of behavioral and subjective measurements. The test results revealed that neither measurement could assume a normal distribution under each experimental condition (context condition (REST vs. DEMO) and dialogue condition (UND vs. MIS)). Thus, we performed the Wilcoxon rank-sum test for behavioral measurements and two-way ordinal regression analyses of variance for subjective measurements.

A. Behavioral Results

The results showed a relatively and significantly low ISA rate compared to that in Williams et al. [4]. In our study, the ISA rate to total utterance was 10.53%, and only 16.7% of participants used at least one ISA utterance, while in Williams et al. [4], the ISA rate to total utterance was 52%, and 69% of participants used at least one ISA utterance. In addition, in our study, the mean of all participants’ ISA rates was 7.93%, whereas in Williams et al. [4], it was 46%. These results support **H1**.

When the independent variable was the dialogue condition and the dependent variable was ISA, a non-significant result was obtained ($W = 311.5$, $p = 0.46$, Wilcoxon rank-sum test; Fig. 3). The median ISA rates in the UND dialogue and MIS dialogue were both as low as 0%. This result supports **H2**. The non-significant result was also found when the independent variable was the context condition and the dependent variable was the ISA rate ($W = 317.5$, $p = 0.36$, Wilcoxon rank-sum

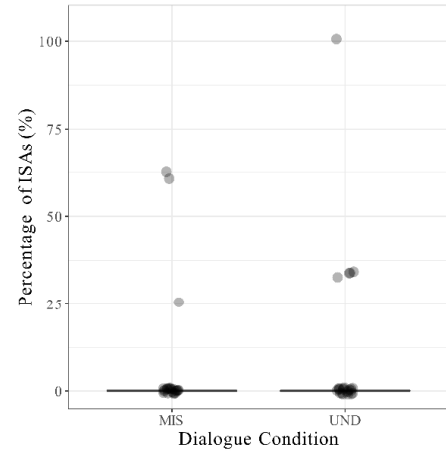


Fig. 3. ISA rates across the dialogue condition

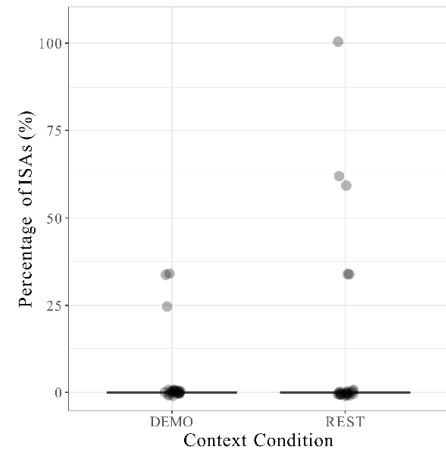


Fig. 4. ISA rates across the context condition

test). As shown in Fig. 4, the ISAs were frequently used in neither the DEMO scenario (0%) nor the REST scenario (0%). This result differed from Williams et al.’s [4] result, where ISA was more frequently used in a conventionalized context (REST scenario), thus supporting **H3**.

In contrast, 28.29% of requests were honorifics, and 27.1% of the participants (13 out of 48) produced at least one honorific. Among the 13 participants who used at least one honorific, 84.6% (11/13) were in the REST scenario. The mean honorific rate of the REST scenario (44.4 ± 49.80) was higher than that of the DEMO scenario (5.21 ± 20.82 ; Fig. 5). The significant difference between both scenarios supports **H4** ($W = 401$, $p < .01$, Wilcoxon rank-sum test).

B. Subjective Results

We performed two-way ordinal regression analyses of variance and Tukey’s HSD test for a post hoc pairwise comparison to analyze the effect of the dialogue condition on the participants’ perception of the robot. The test results are shown in Table IV, which indicate that the Likert scores of all questions for the UND dialogue and the MIS dialogue were not statistically significantly different. These results support **H4**.

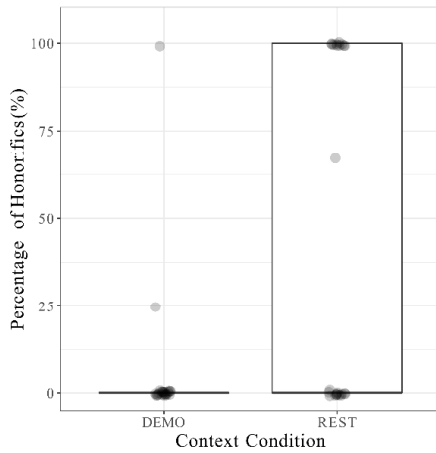


Fig. 5. Honorific rates across the context condition

TABLE IV. EFFECT OF DIALOGUE CONDITION ON PERCEPTION OF ROBOT

	Questions	Z	p	MIS	UND
Q1	The robot was easy to interact with (from 1 to 10, “strongly disagree” to “strongly agree”)	0.92	1	8.17 (± 1.66)	8.71 (± 1.20)
Q2	How would you rate the ease of interacting with the robot? (from 1 to 10, “easy” to “hard”)	-2.69	0.44	4.46 (± 3.26)	2.92 (± 2.80)
Q3	How would you rate the difficulty of the task (from 1 to 10, “easy” to “hard”)	-0.20	1	2.54 (± 2.17)	2.13 (± 1.30)
Q4	The robot was responsive to my commands (from 1 to 10, “strongly disagree” to “strongly agree”)	1.29	0.99	8.92 (± 1.10)	9.08 (± 1.77)
Q5	The robot was cooperative (from 1 to 10, “strongly disagree” to “strongly agree”)	0.18	1	8.88 (± 1.30)	8.96 (± 1.20)
Q6	The robot was helpful (from 1 to 10, “strongly disagree” to “strongly agree”)	-0.13	1	8.58 (± 1.59)	8.71 (± 1.12)
Q7	The robot was annoying (from 1 to 10, “strongly disagree” to “strongly agree”)	-0.08	1	2.29 (± 2.01)	2.04 (± 1.68)
Q8	How would you rate the robot’s level of comprehension? (from 1 to 10 “low” to “high”)	0.76	1	8.00 (± 1.67)	8.33 (± 1.74)
Q9	Did you feel that the robot understood what you were saying? (from 1 to 10, “no” to “yes”)	1.07	1	8.29 (± 1.63)	8.83 (± 1.24)
Q10	Did the robot meet your expectations? (from 1 to 10, “no” to “yes”)	0.00	1	8.25 (± 2.29)	8.50 (± 1.67)

VI. DISCUSSION

We conducted a replication experiment to compare ISA use between English and Korean. The results support most of our hypotheses, suggesting the need for further studies on how to deal with different politeness expressions depending on language.

A. Implications

The results support **H1**, according to which Korean speakers will use ISAs less than English speakers in the same task-based human–robot dialogue as that used by Williams et al. [4]. In their study [4], ISAs accounted for the majority in terms of the number of participants and task-relevant utterances. In contrast, in this study, ISAs constituted to be in minority, which was approximately 10% of the participants and utterances. This suggests that humans’ ISAs for requests are not a severe problem in Korean NLU for simple task-based HRI.

According to **H2**, ISAs will appear at a low rate regardless of the robot’s ability to understand ISAs. The results showed that, in both dialogues, the participants rarely changed their direct requests. This result appears similar to Williams et al.’s [4] result in that there was no difference in ISA rates according to the dialogue condition, and the participants preferred to keep exhibiting the same linguistic behavior. However, this result is different from Williams et al.’s [4] result in that in their study [4], the participants in the MIS dialogue sometimes resistantly continued to use ISAs despite the robot’s failure, while in our study, the participants repeated ISAs only two to four times and changed their utterances into DSAs, showing less aversion toward DSAs. Therefore, in the case of Korean, the need for an NLU system that processes indirect requests or a robot’s ability to induce direct requests from human interlocutors might be relatively low.

According to **H3**, ISAs will appear at a low rate regardless of the conventionality of the tasks. Our result was clearly different from Williams et al.’s [4] result, where the ISAs were more frequent in the conventionalized context. This is consistent with Korean ISA use in HHI illustrated by Byon [7], where the participants produced DSAs more frequently than ISAs in multiple request situations.

According to **H4**, the participants will not perceive the robot differently depending on the dialogue condition. Our result supports this hypothesis in that there was no statistically significant difference between the UND and MIS dialogues in the 10 questions about the perception of the robot. This was contrary to Williams et al.’s [4] result that humans favored the robot unable to understand ISAs less than the robot able to understand ISAs. This result is attributed to the significantly low ISA rate, regardless of the dialogue condition, as mentioned in **H2**. Only three participants experienced the robot’s failure in the MIS dialogue. Even these participants quickly switched their requests into DSAs, getting out of failure.

According to **H5**, the honorific rate will differ depending on the context condition. The results showed that the honorific rate in the REST context was higher than that in the DEMO context. Considering that the context condition did not affect the ISA rate in **H3**, this result suggests that Koreans prefer discernment politeness represented by honorifics over volitional politeness represented by ISAs. That is, in the REST context, Korean speakers are more likely to use honorifics to index social distance from strangers (i.e., waiters), while English speakers are more likely to use ISAs to protect the faces of waiters' faces. However, despite the different politeness expressions and politeness selected from those for English, our results were consistent with Williams et al.'s [4] results, according to which the use of the politeness expression is higher in a conventionalized context than in an unconventionalized context. This result indicates that the conventionalized context has strong constraints on making people follow socio-cultural norms.

B. Limitations and Future Work

The current study provides evidence for the use of different politeness strategies in HRI across languages and reconfirms the effect of the conventionality of the context on the use of politeness strategies in HRI. However, it has many limitations that need to be addressed.

First, although we tried to replicate Williams et al.'s [4] experiment as exactly as possible, except for the language, there were a few differences in terms of the participants' demographics and the experimental materials, especially the robot. The age range of the participants in our study was narrower, from 20s to 30s, compared to that in Williams et al. [4], which was 10s to 60s. People in their 20s and 30s are more accustomed to interacting with agents. Similarly, even though we excluded the applicants majoring in HRI or HCI, our study participants mostly majored in computer science or engineering and had more experience with robots. The results of this experiment, therefore, can be applied differently to people of different ages and robot experiences. In addition, while Williams et al. [4] used iRobot Create, we used Turtlebot. Furthermore, while Williams et al. [4] utilized the robot control architecture (Distributed Integrated Affect, Reflection, and Cognition (DIARC)) [22] to generate the robot's utterances, our study developed the WOZ interface with a human expert (majoring in Korean linguistics) by selecting pre-recorded audio files as the robot's utterances. The possibility that these differences subtly influenced the results cannot be completely excluded.

Second, we shared the same limitation as that of Williams et al. [4] in that our observation was limited to simple tasks with a few contexts. In particular, the results are only applicable in contexts where humans make requests to robots. If the context is not for requesting, different results might appear. In addition, the two contexts alone can be insufficient to answer the question of what makes a context conventional. The study

should be extended to various contexts to investigate the factors of context conventionality.

Our results suggest that cultural differences in HRI need to be examined before constructing an NLU system. In a language using honorifics with morphological differentiation, such as Korean and Japanese, the NLU system that responds flexibly in the presence of honorifics might be more important than the one that estimates the intentions of indirect expressions. According to Reeves and Nass [23], due to the expectation of reciprocity, if people are polite to machines, we need to make machines polite as well. Therefore, if humans use honorifics, it would be desirable for robots to use honorifics as well. In addition, considering discernment politeness in Korean, it is more recommendable for robots to use honorifics. As discussed before, discernment politeness indicates the relative status of interlocutors. As humans are more likely to be perceived as having a higher status than robots, robots should use honorifics to indicate humans' higher status. People in their 20s and 30s generally show a clear preference for smart speakers using honorifics through a perception survey after the interaction between the people and the smart speaker [15]. Thus, it is preferable for robots to use honorifics, even if humans do not use honorifics. Further studies are required to prove this hypothesis.

VII. CONCLUSION

This paper presents the use of ISAs and honorifics of native Korean speakers in task-based HRI. We performed a replication experiment of Williams et al. [4] to illustrate the cultural differences in politeness expressions and politeness between English and Korean in HRI. Our results showed a significantly low frequency of ISAs in Korean, as compared to that in English. Instead, considerable use of honorifics was found in Korean. Moreover, the honorifics were used more in a conventionalized context than an unconventionalized context in Korean, while ISAs were used more in a conventionalized context than an unconventionalized context in English.

Our findings suggest that cultural differences in politeness expressions and politeness manifested in cross-linguistic studies can also be applied to HRI. In particular, the finding that when the context becomes conventionalized, ISA use increases in English and honorific use increases in Korean suggests that the conventionality of context is an important element to consider when investigating linguistic behavior in HRI.

Our research emphasizes the need to further investigate the cultural differences in politeness expressions and politeness in more varied languages and contexts. Given our findings, such cross-cultural studies on HRI will enable NLU system developers or robot designers to develop efficient and cross-culturally adaptable robots and NLU systems.

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